

BEAM SUPPORTED BY ROD HANGERS

MAX + M CONT'D

(w/o cant)

$$+ M_{LL} - w \cdot 60 \times 6^2 / 8 = 270^{1\#} < 450^{1\#}$$

DOES NOT GOV.

$$- M_{DL, \text{CANT}} = 222^{1\#} \text{ SEE PREVIOUS PAGE}$$

$$\begin{aligned} \text{MAX + M WITH DL ON CANT} &= + 300^{1\#} \text{ LL AT CTR SPAN!} \\ &= 649^{1\#} - 222^{1\#} \\ &= 427^{1\#} \text{ LESS THAN MAX CANT MOMENT} \end{aligned}$$

BEARING ON HANGER PLATE WASHERS

EFFECTIVE BRG AREA

$$= 3.5 \times 3.5 - 3.5 \times 0.75$$

$$= 9.6^{in^2}$$

$$f_{brg} = 718^{1\#} / 9.6 = 75^{1\#/in^2}$$

BENDING -

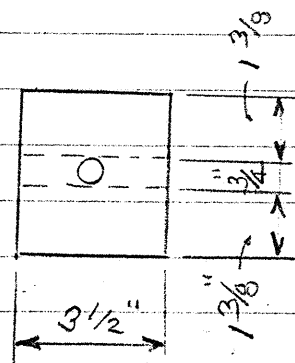
$$\begin{aligned} -M &= (718/2) \times 0.75^{in} \\ &= 269^{in\#} \end{aligned}$$

$$\text{FOR } F_b = 20,000^{1\#/in^2}$$

$$\begin{aligned} \text{REQ'D } S &= 269 / 20,000 \\ &= 0.0135 \text{ IN}^3 \end{aligned}$$

$$1/4" \text{ PL} - 2 3/4" \text{ WIDE}$$

$$\begin{aligned} S &= 2.75 \times 0.25^2 / 6 \\ &= 0.0286 \text{ IN}^3 \end{aligned}$$



5/16 PLATE OK